



4.2.1 The four fundamental spaces of a matrix



04.2.1 The Four Fundamental Spaces

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We assume that the reader remembers theory related to (vector) subspaces. If a review is in order, we suggest Weeks 9 and 10 of Linear Algebra: Foundations to Frontiers (LAFF) [\[27\]](#).

At some point in your linear algebra education, you should also have learned about the four fundamental spaces of a matrix $A \in \mathbb{C}^{m \times n}$ (although perhaps only for the real-valued case):

- The column space, $\mathcal{C}(A)$, which is equal to the set of all vectors that are linear combinations of the columns of A

$$\{y \mid y = Ax\}.$$

- The null space, $\mathcal{N}(A)$, which is equal to the set of all vectors that are mapped to the zero vector by A

$$\{x \mid Ax = 0\}.$$

- The row space, $\mathcal{R}(A)$, which is equal to the set

$$\{y \mid y^H = x^H A\}.$$

Notice that $\mathcal{R}(A) = \mathcal{C}(A^H)$.

- The left null space, which is equal to the set of all vectors

$$\{x \mid x^H A = 0\}.$$

Notice that this set is equal to $\mathcal{N}(A^H)$.

Definition 4.2.1.1. Orthogonal subspaces. Two subspaces $S, T \subset \mathbb{C}^n$ are orthogonal if any two arbitrary vectors (and hence all vectors) $x \in S$ and $y \in T$ are orthogonal: $x^H y = 0$.

The following exercises help you refresh your skills regarding these subspaces.

Homework 4.2.1.1. Let $A \in \mathbb{C}^{m \times n}$. Show that its row space, $\mathcal{R}(A)$, and null space, $\mathcal{N}(A)$, are orthogonal.

▼ [Solution](#)

Pick arbitrary $x \in \mathcal{R}(A)$ and $y \in \mathcal{N}(A)$. We need to show that these two vectors are orthogonal. Then

$$\begin{aligned} x^H y &= \langle x \in \mathcal{R}(A) \text{ iff there exists } z \text{ s. t. } x = A^H z \rangle \\ (A^H z)^H y &= \langle \text{transposition of product} \rangle \\ z^H A y &= \langle y \in \mathcal{N}(A) \rangle \\ z^H 0 &= 0. \end{aligned}$$

Homework 4.2.1.2. Let $A \in \mathbb{C}^{m \times n}$. Show that its column space, $\mathcal{C}(A)$, and left null space, $\mathcal{N}(A^H)$, are orthogonal.

▼ [Solution](#)

Pick arbitrary $x \in \mathcal{C}(A)$ and $y \in \mathcal{N}(A^H)$. Then

$$\begin{aligned} x^H y &= \langle x \in \mathcal{C}(A) \text{ iff there exists } z \text{ s. t. } x = Az \rangle \\ (Az)^H y &= \langle \text{transposition of product} \rangle \\ z^H A^H y &= \langle y \in \mathcal{N}(A^H) \rangle \\ z^H 0 &= 0. \end{aligned}$$

Homework 4.2.1.3. Let $\{s_0, \dots, s_{r-1}\}$ be a basis for subspace $\mathcal{S} \subset \mathbb{C}^n$ and $\{t_0, \dots, t_{k-1}\}$ be a basis for subspace $\mathcal{T} \subset \mathbb{C}^n$. Show that the following are equivalent statements:

- Subspaces $\mathcal{S}, \mathcal{T}$ are orthogonal.
- The vectors in $\{s_0, \dots, s_{r-1}\}$ are orthogonal to the vectors in $\{t_0, \dots, t_{k-1}\}$.
- $s_i^H t_j = 0$ for all $0 \leq i < r$ and $0 \leq j < k$.
- $\left(\begin{array}{c|ccc} s_0 & \cdots & s_{r-1} \end{array} \right)^H \left(\begin{array}{c|ccc} t_0 & \cdots & t_{k-1} \end{array} \right) = 0$, the zero matrix of appropriate size.

▼ [Solution](#)

We are going to prove the equivalence of all the statements by showing that 1. implies 2., 2. implies 3., 3. implies 4., and 4. implies 1.

- 1. implies 2.

Subspaces $\mathcal{S}$ and $\mathcal{T}$ are orthogonal if any vectors $x \in \mathcal{S}$ and $y \in \mathcal{T}$ are orthogonal. Obviously, this means that s_i is orthogonal to t_j for $0 \leq i < r$ and $0 \leq j < k$.

- 2. implies 3.

This is true by definition of what it means for two sets of vectors to be orthogonoal.

- 3. implies 4.

$$\left(\begin{array}{c|ccc} s_0 & \cdots & s_{r-1} \end{array} \right)^H \left(\begin{array}{c|ccc} t_0 & \cdots & t_{k-1} \end{array} \right) = \begin{pmatrix} s_0^H t_0 & s_0^H t_1 & \cdots \\ s_1^H t_0 & s_1^H t_1 & \cdots \\ \vdots & \vdots & \end{pmatrix}$$

- 4. implies 1.

We need to show that if $x \in \mathcal{S}$ and $y \in \mathcal{T}$ then $x^H y = 0$.

Notice that

$$x = \left(\begin{array}{c|ccc} s_0 & \cdots & s_{r-1} \end{array} \right) \begin{pmatrix} \hat{x}_0 \\ \vdots \\ \hat{x}_{r-1} \end{pmatrix} \text{ and } y = \left(\begin{array}{c|ccc} t_0 & \cdots & t_{k-1} \end{array} \right) \begin{pmatrix} \hat{\psi}_0 \\ \vdots \\ \hat{\psi}_{k-1} \end{pmatrix}$$

for appropriate choices of $\hat{x}$ and $\hat{y}$. But then

$$\begin{aligned} x^H y &= \left(\left(\begin{array}{c|ccc} s_0 & \cdots & s_{r-1} \end{array} \right) \begin{pmatrix} \hat{x}_0 \\ \vdots \\ \hat{x}_{r-1} \end{pmatrix} \right)^H \left(\begin{array}{c|ccc} t_0 & \cdots & t_{k-1} \end{array} \right) \begin{pmatrix} \hat{\psi}_0 \\ \vdots \\ \hat{\psi}_{k-1} \end{pmatrix} \\ &= \begin{pmatrix} \hat{x}_0 \\ \vdots \\ \hat{x}_{r-1} \end{pmatrix}^H \underbrace{\left(\begin{array}{c|ccc} s_0 & \cdots & s_{r-1} \end{array} \right)^H \left(\begin{array}{c|ccc} t_0 & \cdots & t_{k-1} \end{array} \right)}_{0_{r \times k}} \begin{pmatrix} \hat{\psi}_0 \\ \vdots \\ \hat{\psi}_{k-1} \end{pmatrix} \\ &= 0 \end{aligned}$$

Homework 4.2.1.4. Let $A \in \mathbb{C}^{m \times n}$. Show that any vector $x \in \mathbb{C}^n$ can be written as $x = x_r + x_n$, where $x_r \in \mathcal{R}(A)$ and $x_n \in \mathcal{N}(A)$, and $x_r^H x_n = 0$.

▼ [Hint](#) ▼ [Answer](#) ▼ [Solution](#)

Let r be the rank of matrix A . In a basic linear algebra course you learned that then the dimension of the row space, $\mathcal{R}(A)$, is r and the dimension of the null space, $\mathcal{N}(A)$, is $n - r$.

Let $\{w_0, \dots, w_{r-1}\}$ be a basis for $\mathcal{R}(A)$ and $\{w_r, \dots, w_{n-1}\}$ be a basis for $\mathcal{N}(A)$. Since we know that these two spaces are orthogonal, we know that $\{w_0, \dots, w_{r-1}\}$ are orthogonal to $\{w_r, \dots, w_{n-1}\}$. Hence $\{w_0, \dots, w_{n-1}\}$ are linearly independent and form a basis for $\mathbb{C}^n$. Thus, there exist coefficients $\{\alpha_0, \dots, \alpha_{n-1}\}$ such that

$$\begin{aligned} x &= \alpha_0 w_0 + \cdots + \alpha_{n-1} w_{n-1} \\ &= \text{< split the summation >} \\ &= \underbrace{\alpha_0 w_0 + \cdots + \alpha_{r-1} w_{r-1}}_{x_r} + \underbrace{\alpha_r w_r + \cdots + \alpha_{n-1} w_{n-1}}_{x_n}. \end{aligned}$$

TRUE

Now prove it!

Let r be the rank of matrix A . In a basic linear algebra course you learned that then the dimension of the row space, $\mathcal{R}(A)$, is r and the dimension of the null space, $\mathcal{N}(A)$, is $n - r$.

Let $\{w_0, \dots, w_{r-1}\}$ be a basis for $\mathcal{R}(A)$ and $\{w_r, \dots, w_{n-1}\}$ be a basis for $\mathcal{N}(A)$.

04.2.1 Visualizing the Four Fundamental Spaces

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Figure 4.2.1.2 captures the insights so far.

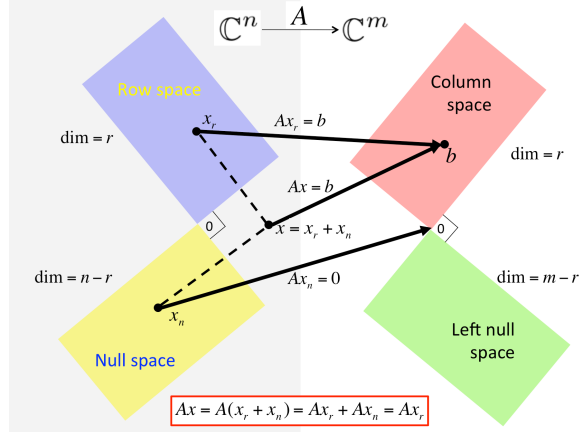


Figure 4.2.1.2. Illustration of the four fundamental spaces and the mapping of a vector $x \in \mathbb{C}^n$ by matrix $A \in \mathbb{C}^{m \times n}$.

That figure also captures that if r is the rank of matrix, then

- $\dim(\mathcal{R}(A)) = \dim(\mathcal{C}(A)) = r$;
- $\dim(\mathcal{N}(A)) = n - r$;
- $\dim(\mathcal{N}(A^H)) = m - r$.

Proving this is a bit cumbersome given the knowledge we have so far, but becomes very easy once we relate the various spaces to the SVD, in

Subsection 4.3.1. So, we just state it for now.